

Report conversation

I'm using python-socketio. Can an "await sio.emit(...)" call be taking time? Or is it strictly fire and forget? Why would it then need to be awaited?

In python-socketio, the `await sio.emit(...)` call can indeed take time. The reason it's an `await` able function and not strictly fire-and-forget is due to its asynchronous nature. Here's a breakdown:

- 1. Networking Overheads:** Even though the emit action itself is quick, it involves networking operations like sending data over a network. These operations are inherently asynchronous and can take variable amounts of time depending on network conditions, server load, etc.
- 2. Waiting for Acknowledgements:** If you're using acknowledgments in your socket.io setup, `emit` will wait for the acknowledgment from the server before completing. This means the function will only complete after receiving a response, which can introduce a delay.
- 3. Concurrency Control:** In async programming, `await` is used to yield control back to the event loop. This allows other asynchronous operations to run while waiting for the `emit` operation to complete. Without `await`, you'd potentially block the execution, which is against the principles of asynchronous programming.
- 4. Exception Handling:** Awaiting the `emit` call allows for proper handling of exceptions that might occur during the transmission. Without `await`, exceptions might not be caught and handled correctly.

In summary, while `sio.emit` is designed to be a quick operation, it's part of an asynchronous system where network latency, acknowledgments, and concurrency control play vital roles, necessitating the use of `await`.